

Attosecond Prediction of Quantum Entanglement
Formation as Supporting Evidence for the
 T_0 -Time-Mass-Duality Theory

Contents

Abstract

This document summarizes the theoretical prediction of the time-resolved formation of quantum entanglement (Jiang et al., 2024) and presents it as supporting evidence for the fundamental time dynamics postulated in the T_0 -Time-Mass-Duality Theory. All theoretical interpretations are drawn exclusively from the content of the Master Narrative (145_FFGFT_donat-teil1_En.pdf) and related documents in the repository: <https://github.com/jpascher/T0-Time-Mass-Duality/tree/main/2/>.

.1 The Theoretical Work

The study by Jiang et al. (2024) demonstrates theoretically that quantum entanglement does **not** form instantaneously, but emerges over a measurable local time window.

Key Details from the Simulation

- **System:** Helium atom driven by an intense high-frequency EUV laser pulse (photoionization).
- **Process:** One electron absorbs energy and escapes (ionizes), while the second electron is excited to a higher energy state.
- **Superposition:** The escaping electron is in a superposition of different departure times (no single sharp instant).
- **Correlation:** The final energy of the bound electron is directly correlated with the departure time of the escaping electron:
 - Higher energy in the bound electron → escaping electron departed earlier
 - Lower energy → escaping electron departed later
- **Predicted Time Window:** Full time-dependent Schrödinger equation simulations yield an entanglement formation window of **~232 attoseconds** ($\approx 2.32 \times 10^{-16}$ s).
- **Proposed Experimental Verification:** A double-pulse scheme (generation pulse + probe pulse) combined with coincidence detection of both electrons to reconstruct the shared quantum history and time the formation process.

Important note: This is a theoretical/numerical prediction. No laboratory measurement has been performed yet. The authors propose a feasible future experiment using current attosecond laser technology.

Popular-Science Video

Video summary of the work: <https://www.youtube.com/watch?v=t3wjY95zvNM> („Scientists Measure Quantum Entanglement Speed — And It Breaks Physics“, Channel: NASA Space News, Uploaded: January 14, 2026)

.2 Connection to the T_0 -Time-Mass-Duality Theory

This theoretical result provides strong conceptual support for the core postulate of the theory:

„In the T_0 -Time-Mass-Duality Theory, time is ontologically equivalent to mass and therefore not merely a coordinate, but an active physical quantity with real dynamics on all scales. Quantum correlations (entanglement) therefore do not arise instantaneously, but develop as a temporal, emergent process within a local interaction window. The predicted attosecond formation time of ~ 232 as confirms exactly this finite, dynamical build-up without non-local ‘spooky action at a distance’ and without violating causality.“

This highlights that all quantum phenomena carry intrinsic time dynamics — a direct consequence of the fundamental duality between time and mass.

.3 References

1. Jiang, W.-C., Zhong, M.-C., Fang, Y.-K., Donsa, S., Březinová, I., Peng, L.-Y., Burgdörfer, J. (2024).
Time Delays as Attosecond Probe of Interelectronic Coherence and Entanglement.
Physical Review Letters **133**, 163201.
DOI: [10.1103/PhysRevLett.133.163201](https://doi.org/10.1103/PhysRevLett.133.163201)
2. Video: „Scientists Measure Quantum Entanglement Speed — And It Breaks Physics“.
YouTube, Channel: NASA Space News.
<https://www.youtube.com/watch?v=t3wjY95zvNM> (accessed January 15, 2026)